

1. Prove or disprove that $\mathbb{Z}$ with Euclidean metric is

- (a) complete
- (b) separable
- (c) compact
- (d) connected

2. Let $X = \{a, b, c, d\}$ and $\mathcal{T} = \{\emptyset, \{a, b\}, \{c\}, \{a, b, c\}, \{a, b, c, d\}\}$.

- (a) Prove that $(X, \mathcal{T})$ is a topological space
- (b) Find the closure of $\{b, c\}$
- (c) Is $\{b\}$ an open subset of $\{b, c\}$ endowed with the relative topology?

3. Prove that the product of two open balls of the same radius in two metric spaces is an open ball in the product metric space.

4. Read the text

For any $n \in \mathbb{N}$ there are open balls $B_{1/n}(x_1^{(n)}), \dots, B_{1/n}(x_{m_n}^{(n)})$ that cover X . Let $S = \{x_k^{(n)}\}_{n \geq 1, 1 \leq k \leq m_n}$. Fix $\varepsilon > 0$ and choose $n \geq \varepsilon^{-1}$. Then for any $x \in X$ there is $x_k^{(n)}$ such that $d(x_k^{(n)}, x) < 1/n$.

- (a) Which statement is proven above?
- (b) Why the last sentence implies the desired result?

5. Read the text

Fix $a \in X$ and consider the sets $B = \{a\} \times Y, C_y = X \times \{y\} \subset X \times Y$ for $y \in Y$. First notice that B and C_y are connected subspaces of $X \times Y$. For B , it follows from the continuity of $f: Y \rightarrow \{a\} \times Y, f(y) = (y, a)$ and Proposition 1.

Now, for each $y \in Y$, we have $C_y \cap B = \{(a, y)\} \neq \emptyset$, and hence the subset $U_y = B \cup C_y$ is connected by Lemma 2. Consider the collection $\{U_y\}_{y \in Y}$. For any $y \neq y' \in Y$, we have $U_y \cap U_{y'} = B \neq \emptyset$, and hence $\cup_{y \in Y} U_y$ is connected by Lemma 2.

- (a) Which statement is proven above?
- (b) Why the last sentence implies the desired result?
- (c) State Proposition 1
- (d) State Lemma 2

6. Let A, B be path-connected subspaces of a metric space X . Prove that $A \cup B$ is path-connected if $A \cap B \neq \emptyset$.

7. Let $\mathcal{T}$ be the collection of subsets of $\mathbb{R}$ which, in addition to $\emptyset$ and $\mathbb{R}$, contains all the subsets whose complements are finite. Prove that $(\mathbb{R}, \mathcal{T})$ is compact.

8. Show that any compact path-connected subset of $\mathbb{R}$ is a closed interval.

9. Is it true that the union of

- (a) a finite
- (b) any

number of complete subsets of a metric space is complete?

10. Let X be a topological space. Prove that the map

$$f: X \rightarrow X \times X, \quad f(x) = (x, x)$$

is continuous.